

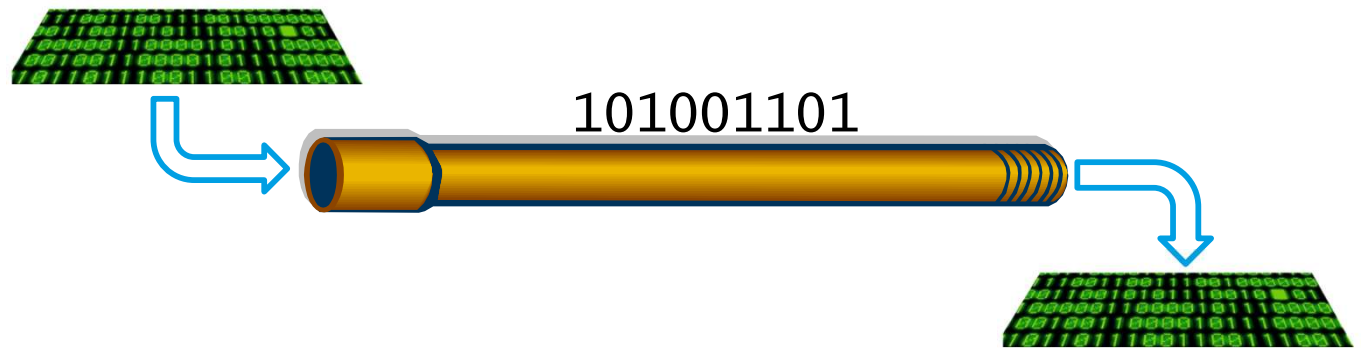
Module 12: Streams

Overview

- Stream
- FileStream
- BinaryReader/BinaryWriter
- StreamReader/StreamWriter
- File/FileInfo
- Directory/DirectoryInfo
- FileSystemWatcher

What is a Stream?

- A generic view of a sequence of bytes.
- Represents an input source or an output destination
- Many different kinds of sources and destinations
 - disk files
 - memory
 - network
 - devices
 - other programs



Namespace System.IO

- Contains types that allow reading and writing to files and data streams and provide basic file and directory support.

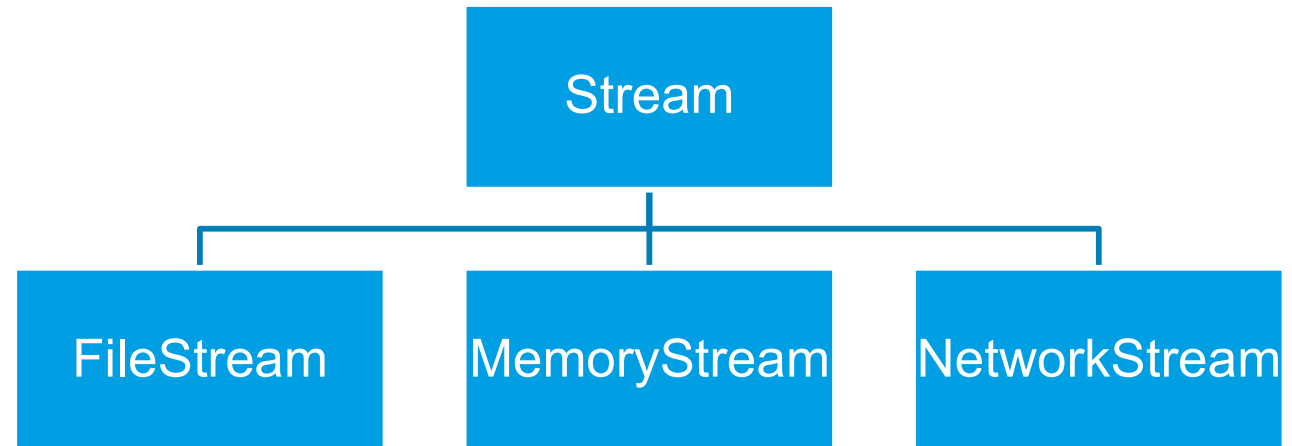


System.IO.Stream

= abstract class

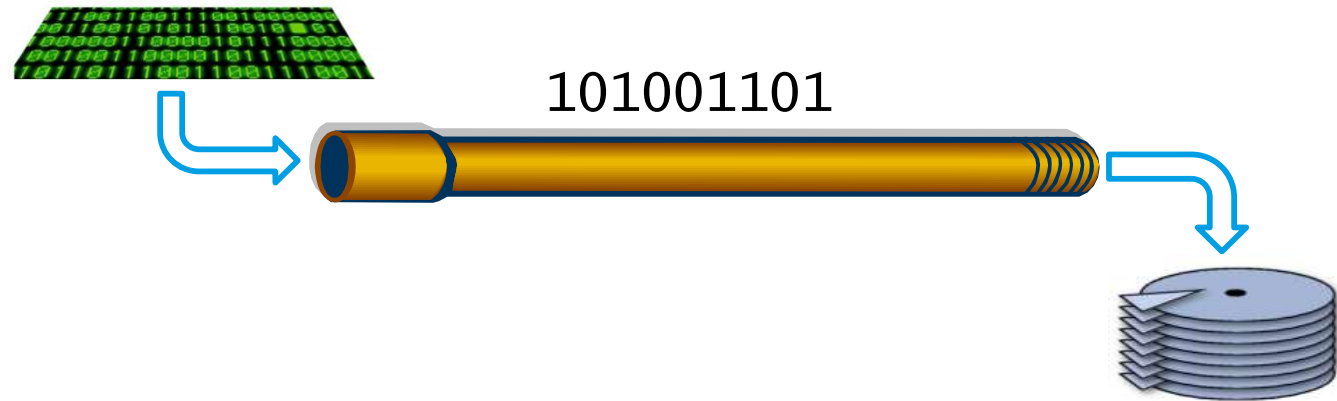
Base functionality of System.IO.Stream

- Close()
- Read()
- Write()
- Flush()
- CanRead
- CanWrite
- CanSeek
- CanTimeout
- etc.



FileStream Class

- to read from, write to, open, and close files

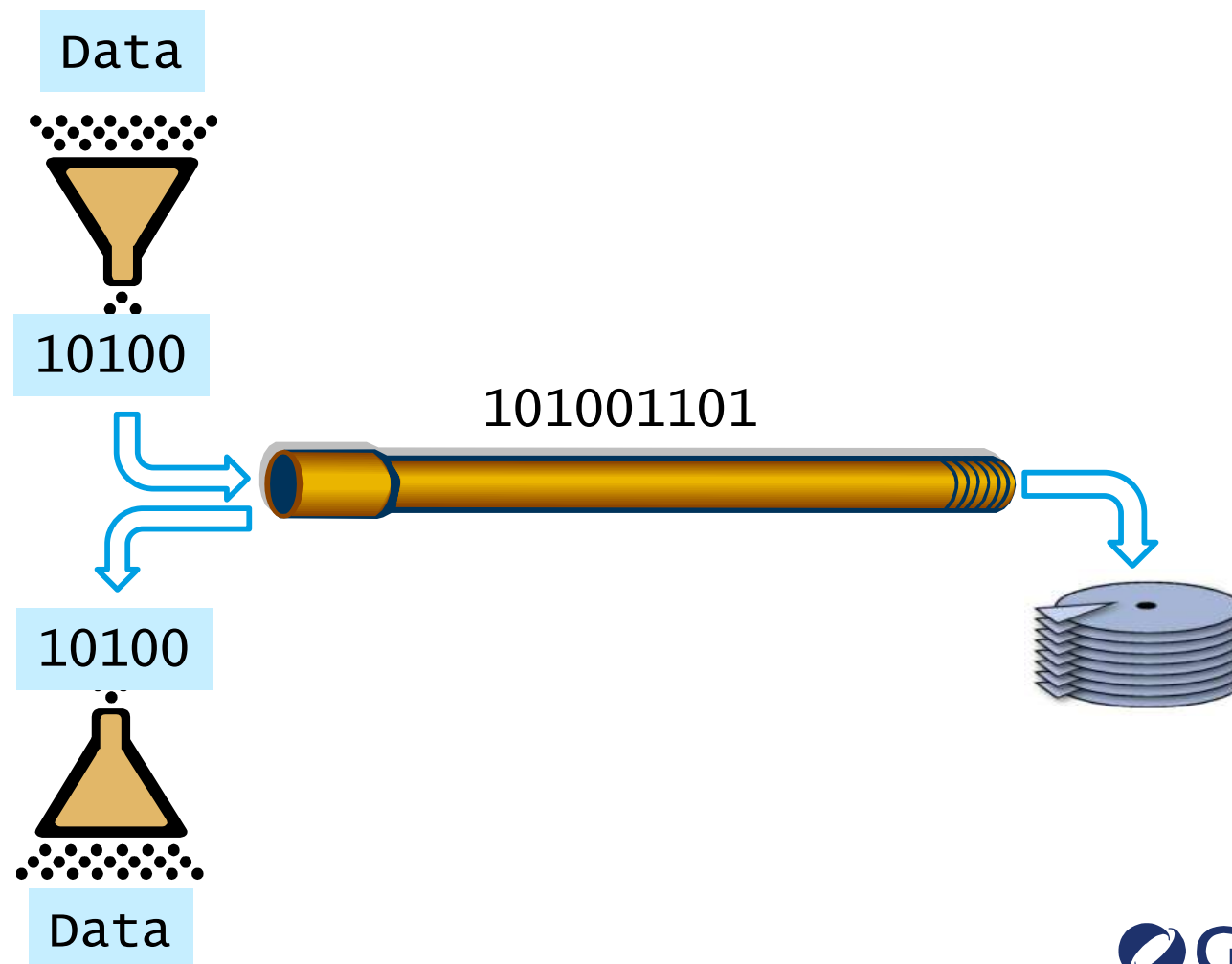


- Writes/reads a block of bytes to the file stream

```
public override void Write(  
    byte[] array,  
    int offset,  
    int count  
)
```

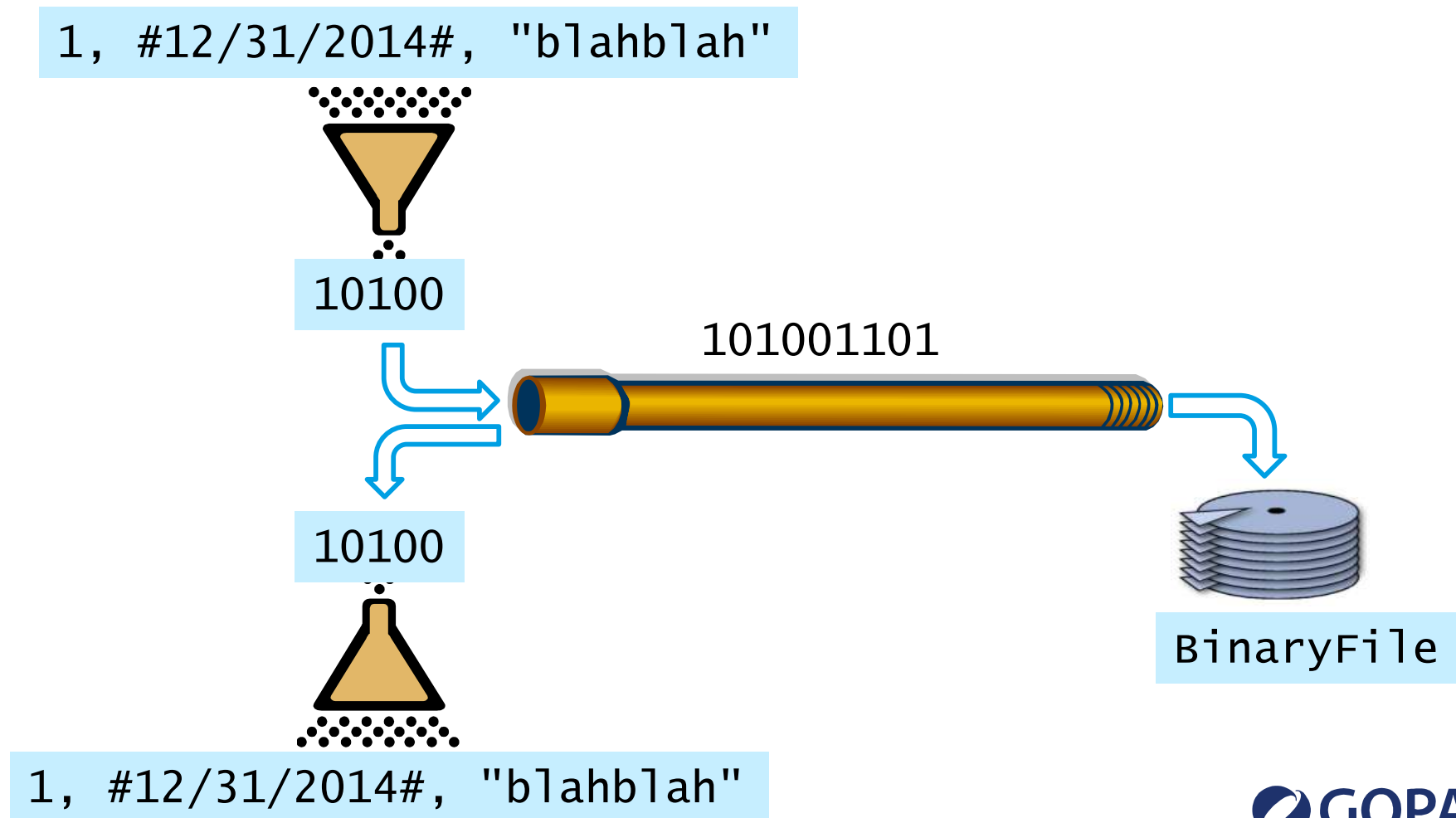
Readers and Writers

- Are responsible for converting things from one format to another



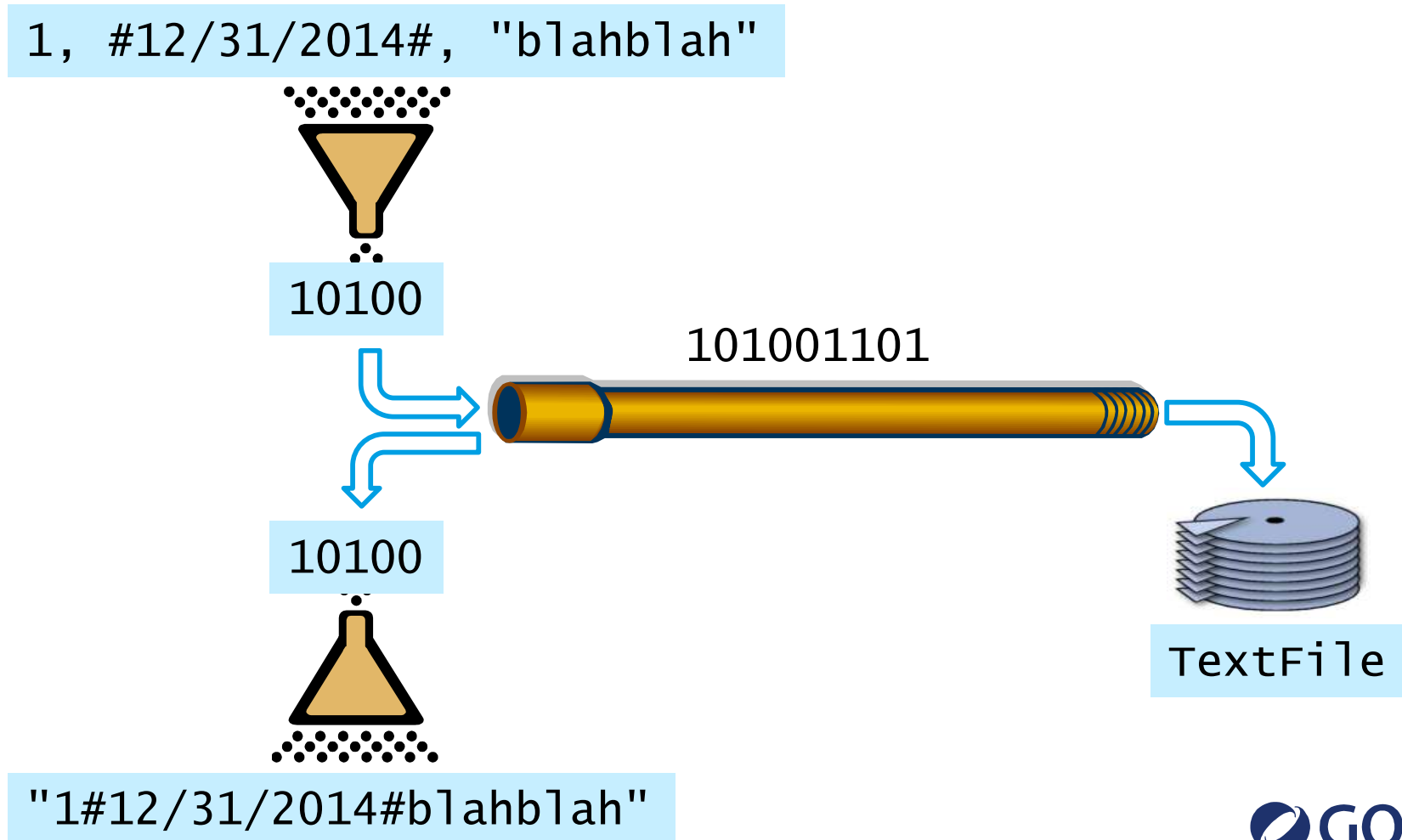
BinaryReader and BinaryWriter

- Reads and Writes primitive data types as binary values



StreamReader and StreamWriter

- Reads and Writes characters from a byte stream in a particular encoding



Text Encoding (UTF-8 encoding by default)

- Use Windows-1250 encoding in .Net Framework

```
using System.Text;
Encoding encoding = Encoding.GetEncoding("Windows-1250");

using (StreamReader sr = new StreamReader(path, encoding))
{
}
```

- Use Windows-1250 encoding in .Net Core

```
using System.Text;
Encoding.RegisterProvider(CodePagesEncodingProvider.Instance);
Encoding encoding = Encoding.GetEncoding("1250");

using (StreamReader sr = new StreamReader(path, encoding))
{
}
```

Lab 12.01 – Streams



Exercise:

1. BinaryReader/ BinaryWriter and FileStream
2. BinaryReader/ BinaryWriter and MemoryStream
3. StreamReader/ StreamWriter a FileStream
4. Encoding

File and FileInfo Class

- File with Static Methods
- FileInfo with Instance

```
FileStream fileStream = File.Create("foo.txt");
```

If you are performing :

- Single operation on a file => File
- Multiple operation on a file => FileInfo

Directory and DirectoryInfo Class

- Directory with Static Methods
- DirectoryInfo with Instance

```
DirectoryInfo dir = new DirectoryInfo(".");  
foreach (FileInfo f in dir.GetFiles("*.cs"))  
{  
    string name = f.FullName;  
}
```

If you are performing :

- Single operation on a directory => use Directory
- Multiple operation on a directory => use DirectoryInfo

Environment Class

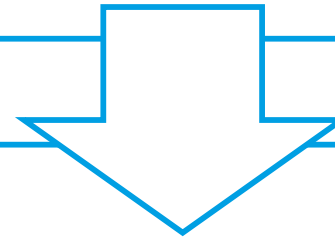
- Provides information about current environment and platform
 - **MachineName** - Gets the NetBIOS name of this local computer
 - **OSVersion** - Gets an OperatingSystem object that contains the current platform identifier and version number.
 - **ProcessorCount** - Gets the number of processors on the current machine.
 - **Version** - Gets a Version object that describes the major, minor, build, and revision numbers of the common language runtime.
 - **NewLine** - Gets the newline string defined for this environment.

Environment.CurrentDirectory

= directory where the process starts

- Gets or sets the fully qualified path of the current working directory.

```
Environment.CurrentDirectory = @"C:\Users\Student\Documents";
```



```
Environment.CurrentDirectory =  
    Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments);
```

Tip: Using Paths

- Accessing environmental variables

```
string myDocPath =  
    Environment.GetEnvironmentVariable("windir");
```

- Combining Paths

```
System.IO.Path.Combine(myDocPath, @"MyApp\hello.txt");  
//C:\Users\Student\Documents\MyApp\hello.txt
```


FileSystemWatcher

- FileSystemWatcher Is Used to Monitor a File System

```
FileSystemWatcher watcher = new FileSystemWatcher();  
  
watcher.Path = @"D:\";  
watcher.Filter = "*.txt";  
watcher.NotifyFilter = NotifyFilters.FileName;  
watcher.Renamed += OnRenamed;  
  
watcher.EnableRaisingEvents = true;
```

Lab 12.02 – File, FileInfo, Directory, DirectoryInfo a FileSystemWatcher



Exercise:

1. File
2. FileInfo
3. Directory
4. DirectoryInfo
5. FileSystemWatcher

Review

- Stream
- FileStream
- BinaryReader/BinaryWriter
- StreamReader/StreamWriter
- File/FileInfo
- Directory/DirectoryInfo
- FileSystemWatcher